

STATIC INCENTIVES → COLLECTIVE KNOWLEDGE

Better advice, weaker learning?

AI, Human Cognition and Knowledge Collapse

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github.com/gsaco/ai-04-acemoglu

Five-minute discussion. Focus: static problem and Observation 1.

1 The paper and the agent's problem

Can better individual advice weaken the knowledge everyone needs?

$$\max_{e \geq 0} U(e; X, \tau_A) = G(X)\Delta_G + G(X)G(Y)\Delta_X - \frac{e^\alpha}{\alpha}, \quad Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

- X : inherited public precision; Y : context-specific precision.
- $G(z) = 2\Phi(\sqrt{z}) - 1$: probability the prediction error is within 1.
- Effort produces private information and a public spillover; short-lived, atomistic agents take X as given.

Production restriction: $\Delta_I = 0$, $\Delta_X > 0$, $\Delta_G + \Delta_X = 1$.

Source: §§3.1–3.3, pp. 7–12, eqs. (4)–(6); constant $f(0, 0)$ omitted.

The supplied paper's §2 is related literature; its static model starts in §3.

Acemoglu, Kong & Ozdaglar (2026)

2 Observation 1: two signs, explicit conditions

$$g(z) = \frac{\phi(\sqrt{z})}{\sqrt{z}}, \quad g'(z) = -\frac{1}{2}(1 + z^{-1})g(z) < 0.$$

$$\underbrace{U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0}_{\text{general knowledge complements effort}}$$

$$\underbrace{U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0}_{\text{agentic advice substitutes for effort}}$$

Conditions: $X > 0$, finite $\tau_A \geq 0$, $\sigma^{-2} > 0$, $\lambda_I > 0$, $\alpha > 1$; Assumption 1, independent Gaussian information, unit-tolerance payoffs; public learning is not privately internalized.

$U_{ee} < 0$ gives a unique interior optimum: $e_X^* > 0$, $e_{\tau_A}^* < 0$. At $X = 0$, $e^* = 0$; the strict interior statements do not apply.

Welfare caution: fixed- X utility rises; long-run welfare need not. The single-peak claims require $\sigma^{-2} \geq \sqrt{2} - 1$ and the relevant $\alpha - 1 \gtrless 1/4$ regime; the optimum may be $\tau_A^* = 0$.

Source: Observation 1, p. 13; Observation 2, p. 14; Propositions 10–11, p. 26.

3 What I did: verify signs, test the hidden premise

Lean checked 10 static results, with no proof holes or added axioms.

Verify the derivation

From the Gaussian integral: G' , g' , the FOC, payoff curvature and both cross-partials.

$$U_{eX} > 0, \quad U_{e\tau_A} < 0.$$

The signs survive with $\Delta_I > 0$ for $X > 0$ under the stated assumptions.

Scope: static proofs only; Propositions 1–16 remain unformalized.

Lean 4 + mathlib; fresh build and axiom audit: 14 September 2026.

Proofs: [lean/ProofInterface.lean](#). Results and full-paper coverage: [lean/audit/](#).

Acemoglu, Kong & Ozdaglar (2026)

Test the hidden premise

Restore standalone value: $\Delta_I G(Y)$. At $X = e = 0$,

$$U_e = \Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0.$$

For $\Delta_I > 0$ and finite AI precision, Lean proves that some $e > 0$ improves on zero effort.

4 Where I challenged the AI's reasoning

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$$Y = \sigma^2 + \lambda_I e + \tau_A, \quad G(z) = 2\Phi(\sqrt{z}) - 1$$

$$g(z) = G'(z) = 2\phi(\sqrt{z}) \cdot \frac{1}{2\sqrt{z}} = \frac{\phi(\sqrt{z})}{\sqrt{z}}$$

$$\log g(z) = -\frac{z}{2} - \log \frac{3}{2} - \log(\sqrt{2\pi})/2$$

$$g'(z) = -\frac{1}{2} \left(\frac{1+\frac{1}{2}}{2} \right) g(z) < 0$$

Con $\Delta_I = 0 \rightarrow \alpha > 1$:

$$U = \Delta_I G(X) + \Delta_X G(X)G(Y) - \frac{e^{\alpha}}{\alpha}$$

$$Y = \lambda_I \rightarrow U_e = \Delta_X G(X) \lambda_I g(Y) - e^{\alpha-1} = 0$$

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0$$

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g(Y) < 0$$

Claim tested: “More accurate AI must raise welfare.”

Verdict: true for optimized static utility at fixed X ; not a general long-run conclusion. Public-knowledge losses also matter.

The assumption audit: §5 relaxes aggregation, synthetic data and effort bundling. It retains $\Delta_I = 0$.

$$U_e|_{X=0, e=0} = \Delta_I \lambda_I g(\sigma^{-2} + \tau_A) > 0$$

My extension: with $\Delta_I > 0$, private learning remains valuable at $X = 0$.

Source: Assumption 1, p. 8; §4.3, pp. 24–26; §5, pp. 29–32.